Alejandro Jaimes

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Education

University of Central Florida — Orlando, FL
Bachelor of Science in Computer Science — *May 2028*

Experience

Software Engineer Intern — May 2026 - Present
Institute for Simulation and Training — *Orlando, FL*

- Constructed a real-time spatial anchoring system using **AR Foundation**, enabling **8** users within **5cm** accuracy.
- Developed networked player-tracker with **Photon**, matching data for **XR** devices at low latency for live demos.
- Implemented runtime **XR subsystem** detection for hardware-aware kiosk mode, reducing demo setup by **100%**.
- Designed backward-compatible multiplayer integration system, maintaining **100%** single-player functionality while adding new multiplayer features across **2** distinct codepaths.

President — Apr 2026 - Present
Graphics Programming Knights — *Orlando, FL*

- Established Graphics Programming Knights, a non-profit student organization, growing membership to **165+**.
- Organized the first community-wide Render Jam competition resulting in **40+** participants and **5** projects.
- Orchestrated **2 workshops per term** on pathways into graphics programming, covering essential **mathematics**, **game development engines**, and **GPU programming** with student attendance of **20+**.
- Expanded club reach by **40%**, establishing partnerships to have workshops and hackathons with **3** organizations.

Outreach Board — Jan 2026 - Present
Knight Hacks — *Orlando, FL*

- Boosted online community engagement online with strategic content creation and event awareness by **57%**.
- Produced and edited Reels, increasing Knight Hacks' visibility and follower engagement, averaging **5k+** views.
- Expanded event attendance by **15%** compared to previous semesters through targeted outreach campaigns on social media and speaking to the public, increasing participation with an average of **40+** people per meeting.

Projects

greghouse | *C++, Arduino, ESP32, React, AWS, Node.js, Lambda, DynamoDB*

- **3rd place winner** of the **Best Overall** category at BloomKnights out of **86** projects and **240+** participants.
- Engineered **ESP32** firmware in **C++** streaming soil telemetry to **AWS** every 2s with sub-3s latency.
- Architected an IoT pipeline with **API Gateway**, **Lambda**, and **DynamoDB** handling **1,800+** readings hourly.
- Crafted a pixel-art **TFT** display on **C++** with mood-driven animations, synchronizing with the **React** app.

sFun (SNES emulator) | *C++, C, CMake, Assembly, SDL2*

- Designed and implemented a **65c816 CPU** emulator supporting, dynamic register width handling using **C++**.
- Built an extensible opcode dispatch system using function pointer tables and implemented core CPU instructions (LDA, REP/SEP, branching, stack operations, JSR/RTS) with correct flag behavior.
- Developed a memory subsystem for SNES RAM/ROM access, enabling instruction fetch, decoding, and execution.
- Engineered robust control flow correctness for relative branching, subroutine calls, and stack-based return handling ensuring **100%** correctness on test cases.

Next Step | *React, Tailwind, Next.js, TypeScript, Python, Flask, Vapi*

- **3rd place winner** of State Farm's Sponsor Challenge at Shell Hacks 2025 out of **50** projects submitted.
- Created an AI voice agent delivering interactive, personalized feedback across **3** different insurance categories.
- Implemented backend services to process and structure conversational data generating dynamic agent responses, reducing response time by **66%** using **Python**, integrated with the **Vapi API**.
- Engineered **7** interactive components for optimal experience with **TypeScript**, **Next.js**, and **React** for frontend.

Technical Skills

Languages: C, C++, C#, Python, HTML, CSS, JavaScript, TypeScript, Assembly, Java
Frameworks: Unity, Next.js, Node.js, Flask, Unreal Engine, Three.js, OpenCV, Mediapipe, CMake, OpenGL, CUDA
Libraries: Tailwind CSS, React, SDL2, WebSockets, TFT_eSPI, Arduino
Tools: ESP32, Docker, AWS, Cloudflare, Git, Arduino IDE, PlatformIO, Vite